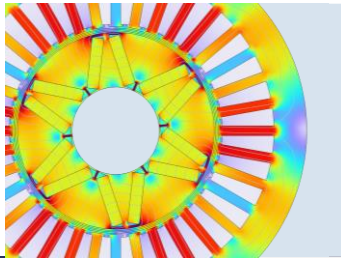


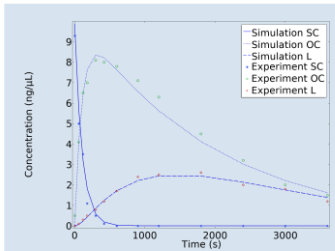
Topology Optimization with COMSOL Multiphysics®

Optimization Capabilities



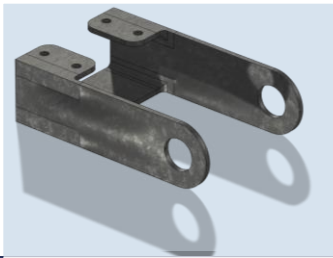
Parameter Optimization

Find the optimal dimensions and position so that they minimize an objective function given a set of constraints.



Parameter Estimation

Find a set of parameter values that minimizes the discrepancy between model and measurement (objective).



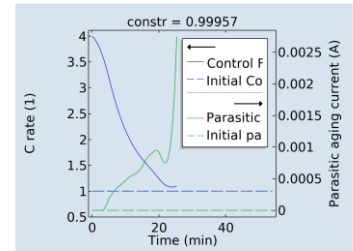
Shape Optimization

Find the optimal shape so that it minimizes an objective function given a set of constraints.



Topology Optimization

Optimize the material distribution to minimize an objective function given a set of constraints.



Optimal Control

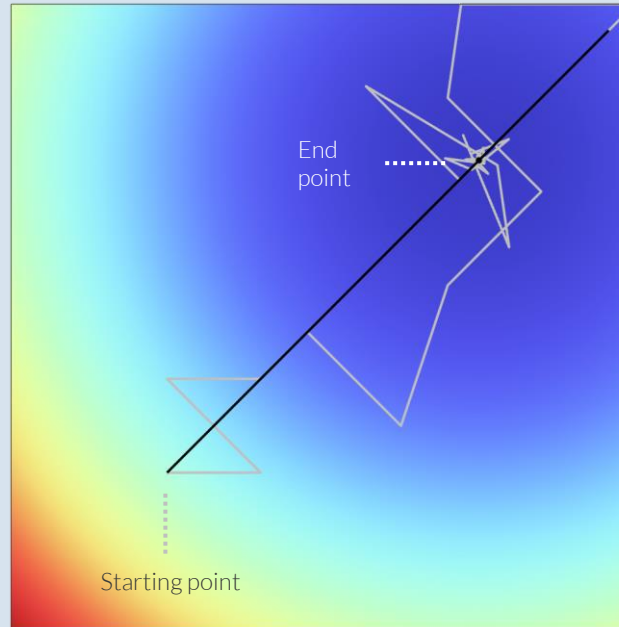
Minimize the time for a process or optimize the control of some other time-dependent problem.

Core Technology

Key characteristics:

COMSOL Multiphysics® has a flexible user interface thanks to the integrated nature of the software. The core technology relies on symbolic differentiation, and this allows for:

- Multiphysics analysis
- Gradient-based optimization



Gradient-Free Optimization

- Nelder–Mead
- COBYLA¹
- BOBYQA²

Gradient-Based Optimization

- GCMMA³/MMA (1st order)
- IPOPT⁴ (2nd order)
- Levenberg–Marquardt (parameter estimation)

¹COBYLA = Constraint Optimization by Linear Approximation

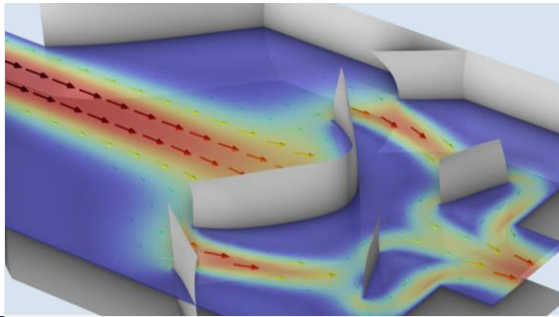
²BOBYQA = Bound Optimization by Quadratic Approximation

³GCMMA = globally convergent method of moving asymptotes

⁴IPOPT = interior point optimizer

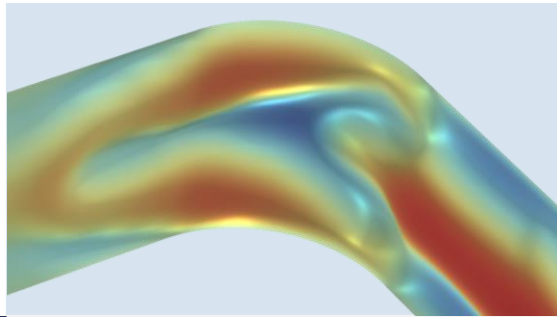
Shape Optimization

Shape Optimization



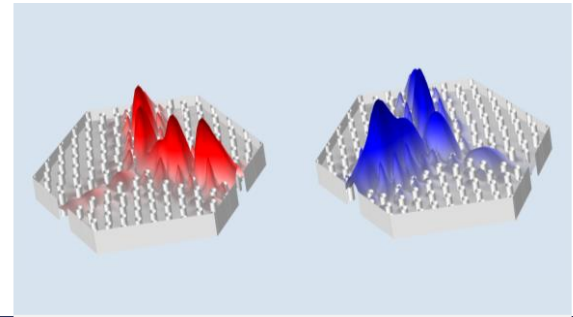
Solids (Volumes)

Get substantial improvements in performance by modifying the geometry, subject to constraints on design freedom.



Shells (Surfaces)

Shape optimization can be applied on surface geometries without a volume.



Shape Preservation

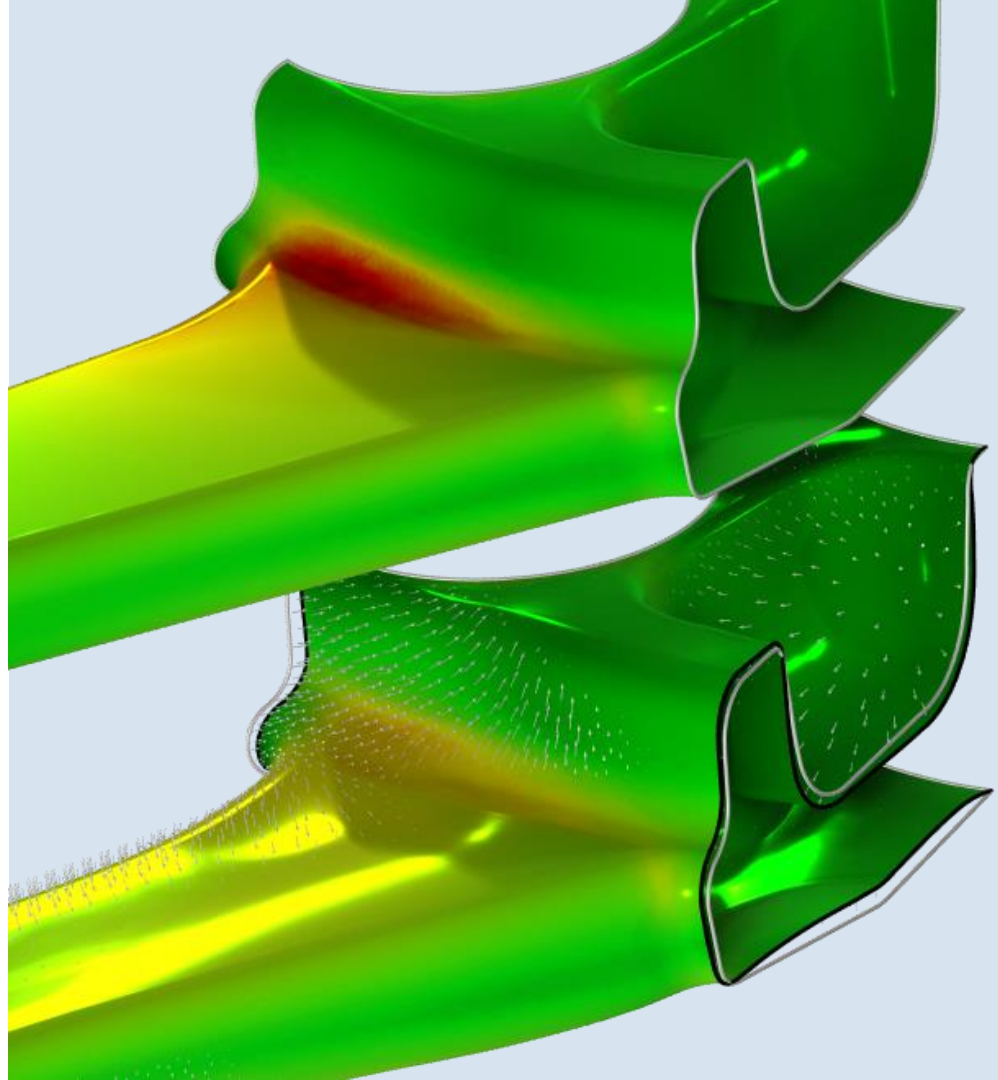
Shape changes — restricted to translation, scaling, and rotation — avoid the high costs associated with parameter optimization.

Adjusting the Design Freedom

Main features:

- PDE¹-based regularization:
 - Preservation of normal continuity
- Polynomial-based regularization:
 - PDE-based parameterization in 3D
 - Normal continuity cannot be preserved in 3D
- Transformation:
 - Translation
 - Scaling
 - Rotation

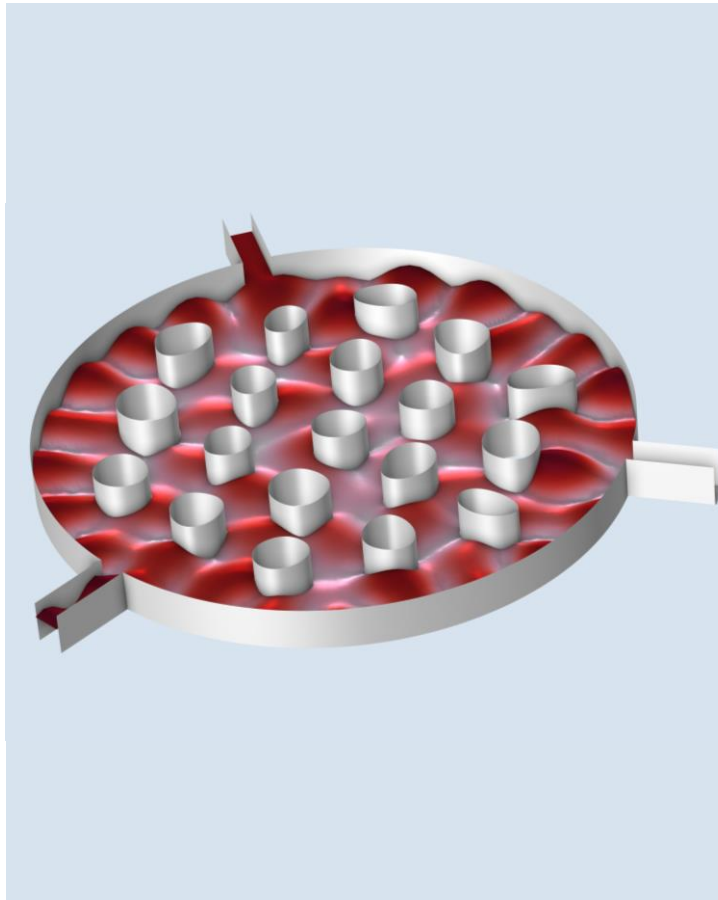
¹PDE = partial differential equation



Free Shape Boundary

A PDE-based low-pass filter is applied to each coordinate so that the deformation of the geometry is smooth and the quality of the mesh is preserved.

The *Free Shape Shell* feature uses the same principle.



Filter Length

The filter removes small features, creating a smooth surface for the obstacles.

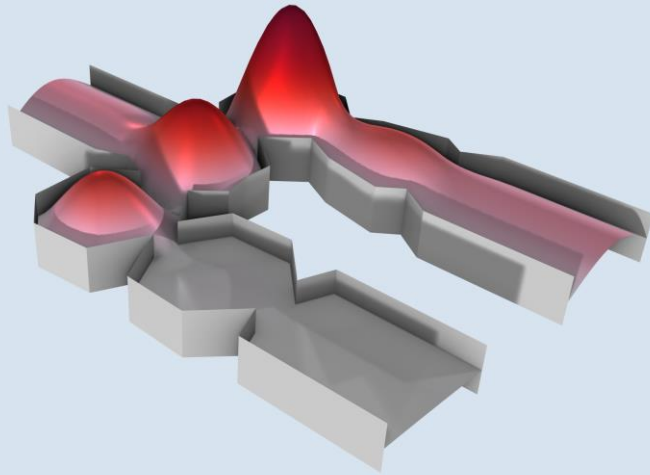
2D vs. 3D

The features work the same in 2D and 3D. They do not rely on a parametric geometry.

Polynomial Boundary

The deformation is defined as an explicit polynomial expression.

Typically, Bezier curves or surfaces are used.



Order

The positioning of the vertices of the geometry are optimized using linear polynomials.

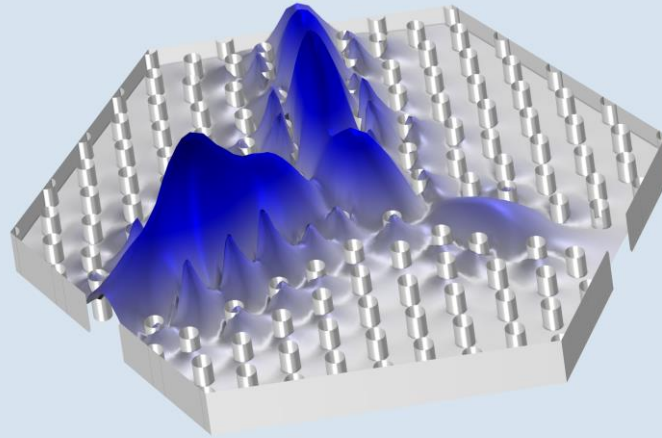
2D vs. 3D

Only triangles and quadrilaterals are supported in 3D. Continuity of the normal is only supported in 2D.

Transformation

The shape deformation is limited to:

- Translation
- Scaling:
 - Isotropic
 - Anisotropic
- Rotation



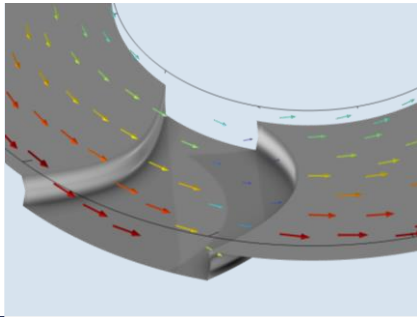
Translation

The movement can be free or limited to a certain plane or axis.

Rotation

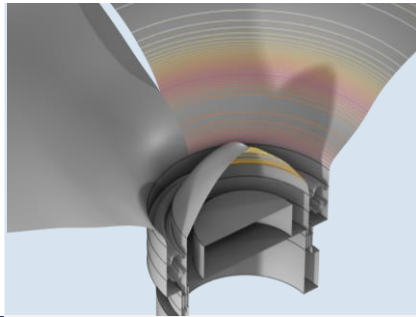
The rotation can be free, or the axis can be limited to a plane or line.

Shape Optimization Example Models



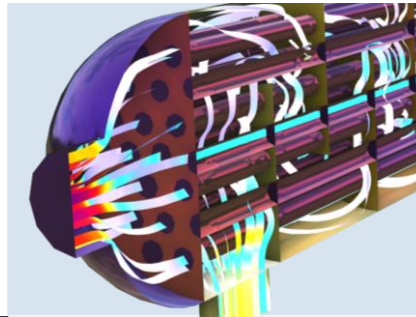
Rotor Dynamics

Maximize bearing capacity by changing the geometry.



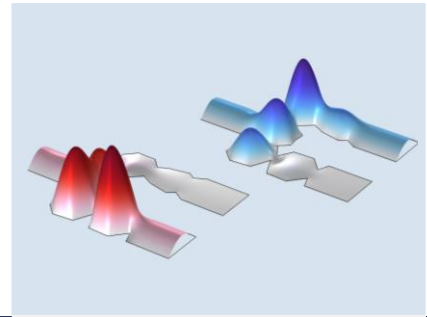
Acoustics

Minimize response variation across frequencies and angles for a loudspeaker tweeter.



Heat Transfer/CFD

Maximize the heat-transfer rate in a heat exchanger with a fixed pressure drop and shaping of tubes.



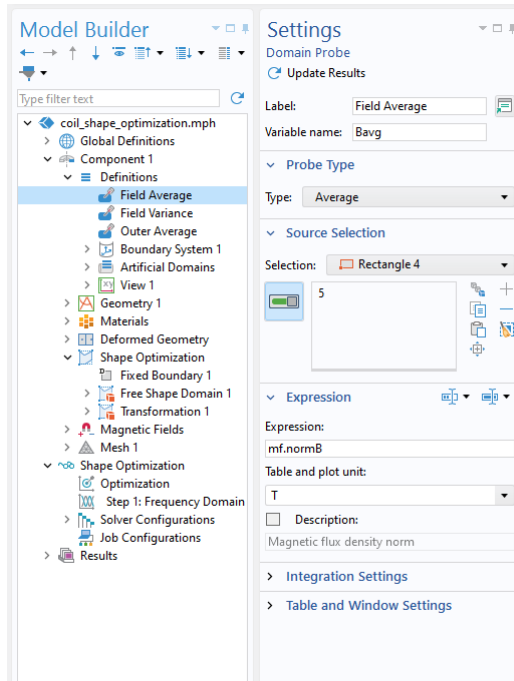
Electromagnetics

Design an RF diplexer by routing different frequencies to different outputs. Reject other frequencies.

EXAMPLE: AC/DC SHAPE OPTIMIZATION Coils

Custom Objective:

Minimize variance of field strength to ensure uniform inductive heating using gradient-based optimization.



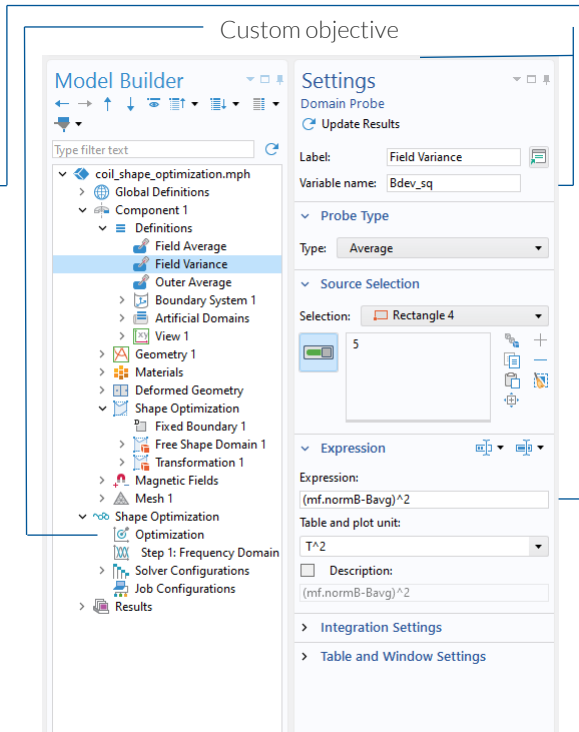
Custom variable

List of variables

Field variable

Compute Average

Use 1st *Domain Probe* to compute average field strength.



Custom objective

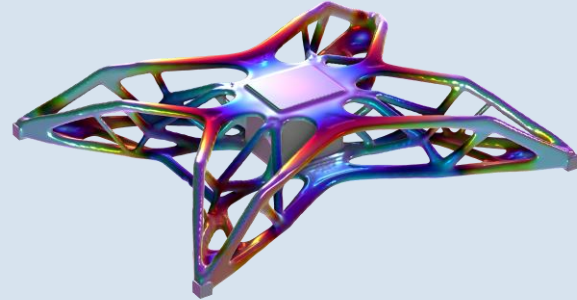
Compute Variance

Use 1st variable in 2nd *Domain Probe* to compute field variance. Use 2nd variable in *Optimization* study step.

Topology Optimization

Topology Optimization

- Optimizes material layout within a given design space for a given set of loads, boundary conditions, and constraints
- Used early in the design process, typically at the conceptual stage
- Treats the distribution of material as a design variable and inserts or removes material to improve the design (objective function)
- Due to the high number of design variables, only gradient-based optimization is practical.



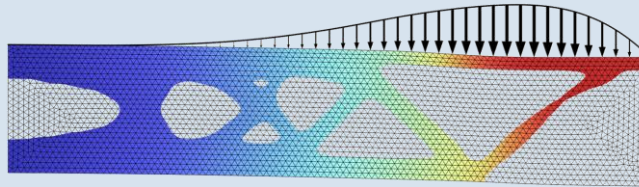
Topology optimization of a drone structure.

Topology Optimization

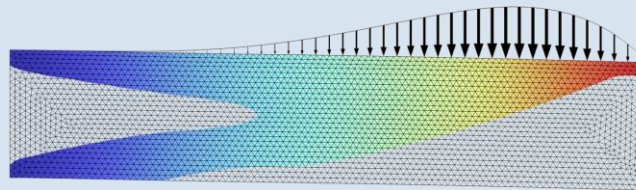
Basic functionality:

- Geometry is defined using spatially varying materials:
 - *Density Model* feature
 - Control variable fields
- Verify performance by exporting to a new component using a *Filter* dataset.

Without milling constraints



With milling constraints



Advantages

- Extreme design freedom yields extreme performance.
- No initial design is required.

Disadvantages

- Settings are specific to the physics. (Multiphysics can require custom interpolation.)
- Slow; many iterations are often required to find a good minimum.

Key Implementation Aspects for Topology Optimization



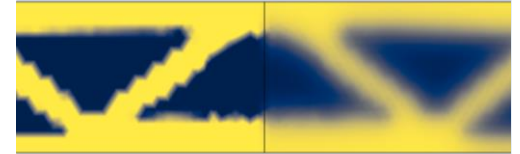
Checkerboarding

- Control variable field: values at each point
- Avoid by filtering



Mesh Dependency

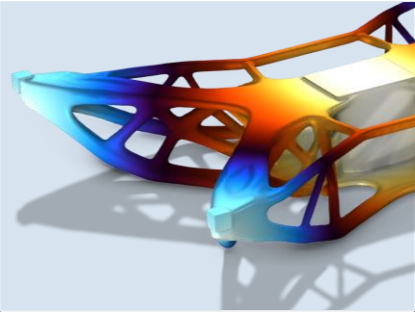
- Mesh density determines feature size
- Avoid by filtering: filter width $\sim$ feature size



Material Interpolation

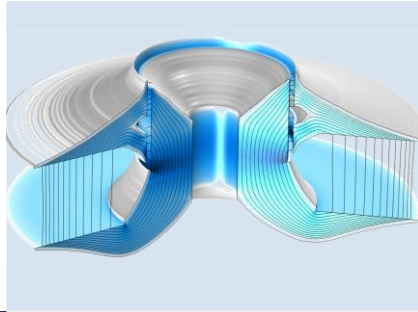
- Smoothed binary material properties
- Avoid numerical issues at material transitions and voids

Topology Optimization: Example Models



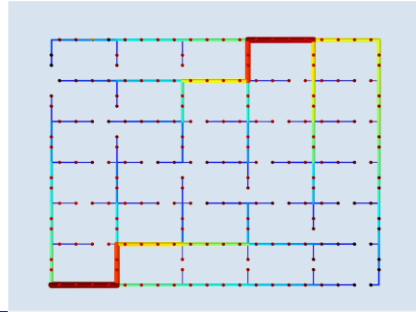
Structural Mechanics

Classic volume-constrained compliance minimization



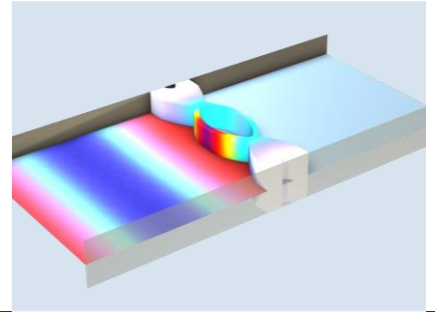
AC/DC

Minimize distortion by optimizing for a flat BL curve.



Heat Transfer/CFD

Optimize district heating network in terms of costs of installation and running.



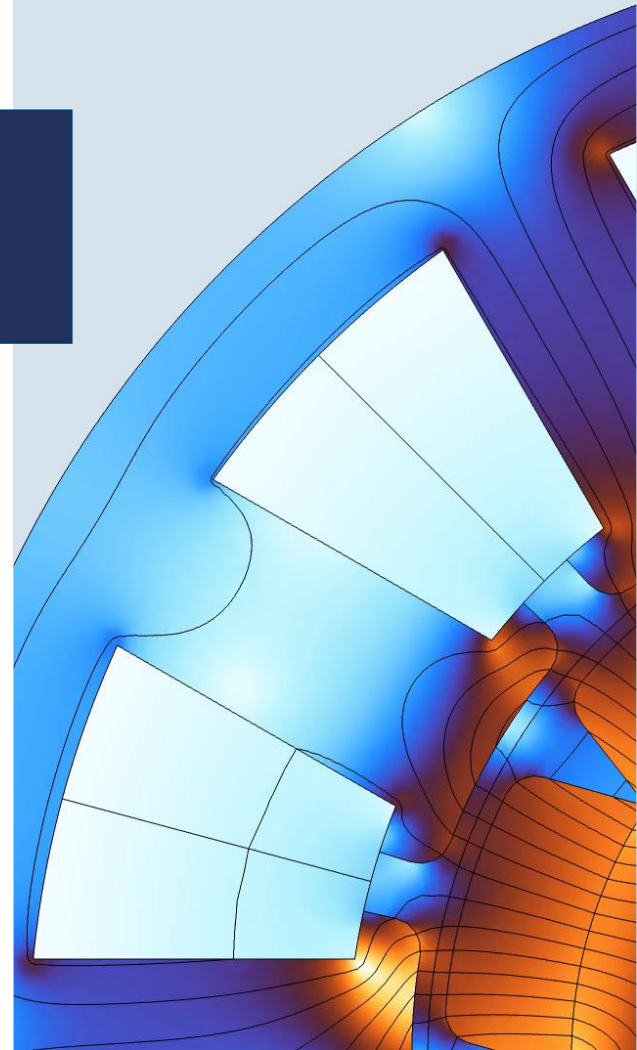
Acoustics

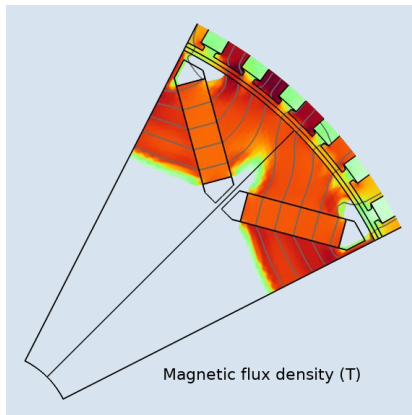
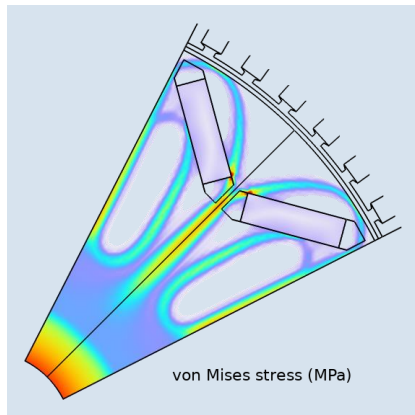
Design sound barrier subject to material constraint.

Time-Periodic Electromagnetics

A Game-Changing Approach

- For nonlinear time-periodic problems:
 - Solves for the steady-state conditions directly
 - Imposes periodicity in the time dimension
 - Solves for all time frames at once
- Supports the exact modeling of rotations, nonlinear magnetic materials, induced currents, and general periodic excitations
- Particularly well suited for:
 - Parametric studies — efficiency maps and temperature maps
 - Multiphysics — heat transfer, deformations, and vibrations
 - Optimization — topology optimization and objectives based on the fast Fourier transform (FFT)
 - Models that require many cycles to reach a steady state





TIME-PERIODIC ANALYSIS

Optimization Strategies

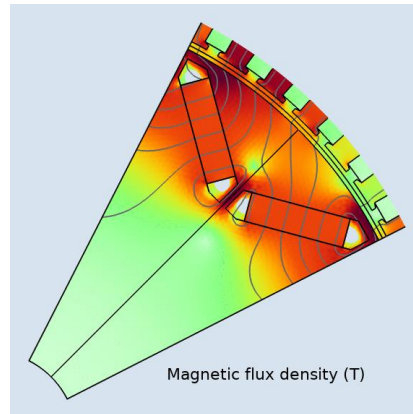
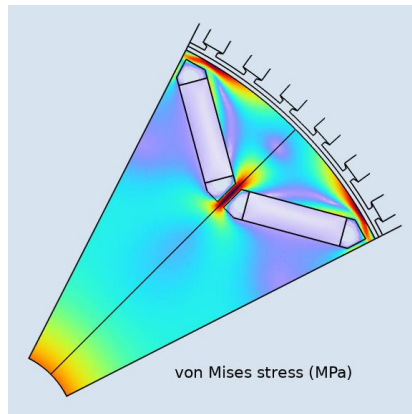
Compatible with all optimization algorithms and multiphysics couplings

Verified final design against starting point: +8% average torque, factor 2.5 reduction in torque fluctuations, -38% iron, and -14% maximum stress

Competing Physics

Above: Structural and magnetic optima lead to a different topology.

Right: multiphysics optimum for stress and magnetic flux simultaneously



LIVE SHOWCASE

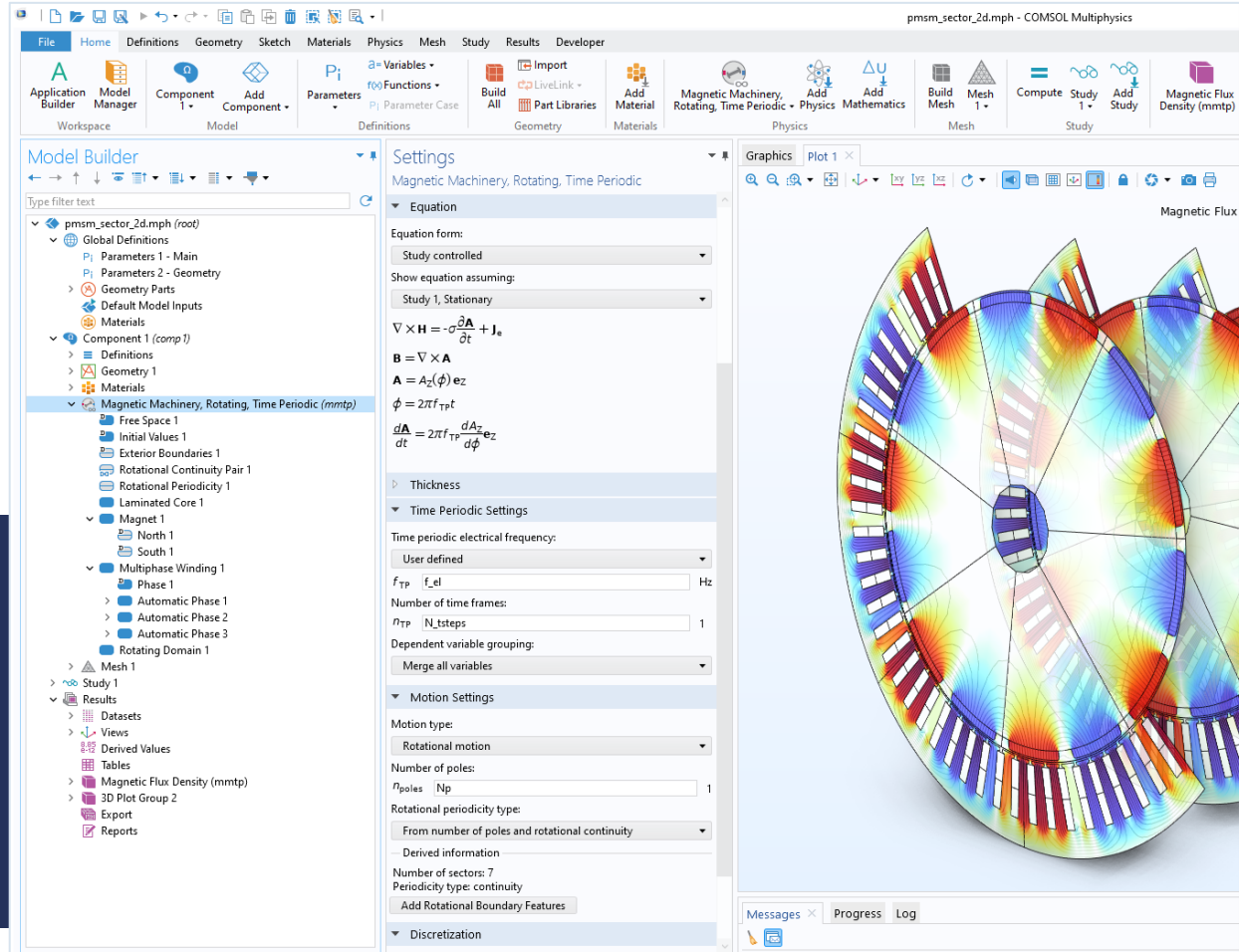
Time-Periodic Analysis and Optimization

Single-Physics Objectives

- Maximum stiffness-to-weight ratio (centrifugal forces)
- Optimal magnetic circuit for torque production

Multiphysics Optimization

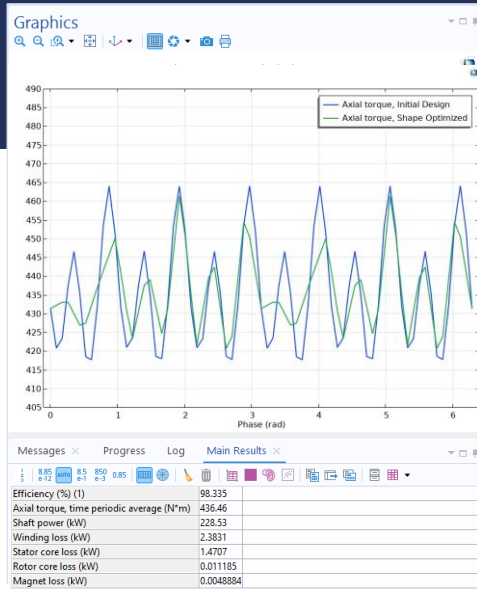
- Combining several objectives
- Results in more realistic design



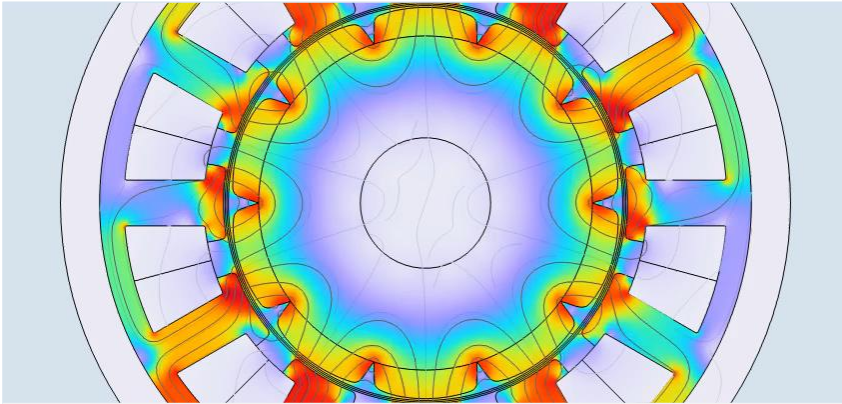
Shape Optimization of Rotor Surface

With gradient-based optimization, free or prescribed deformation of boundaries is effectively formulated as a set of control variables.

Here, a major improvement of the torque ripple is achieved through optimization of the rotor surface.

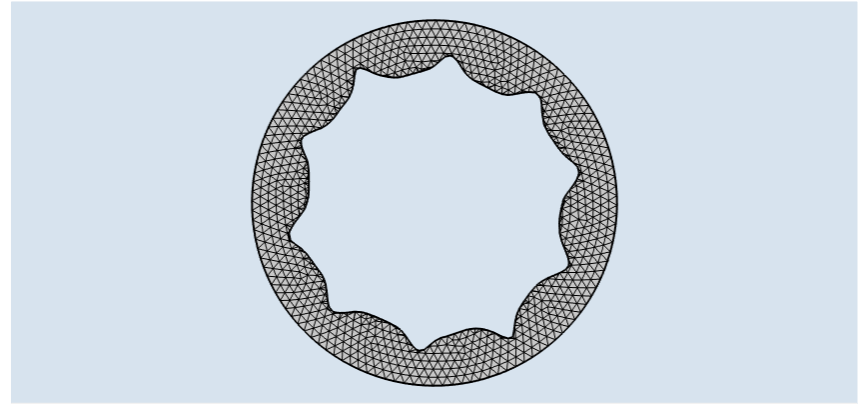


Topology Optimization of the Permanent Magnet Rotor Core



Rotor Core Optimization

Topology optimization determines torque is least sensitive to permeability near the shaft and removes material in this area.



Core with Reduced Weight

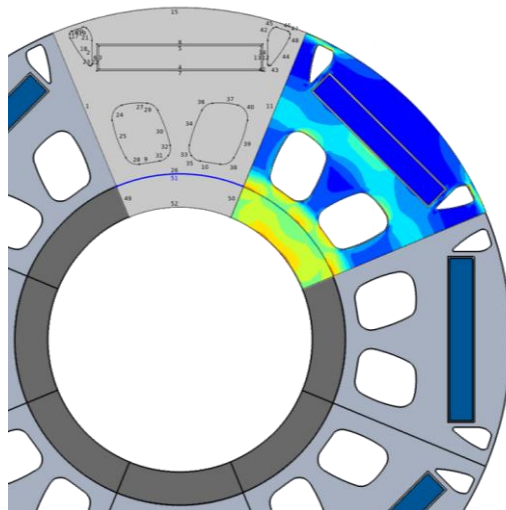
The *theta* field can be converted into a geometry part which can be used for verification or export to a CAD file.

INNOVATION AT
Volkswagen Kassel

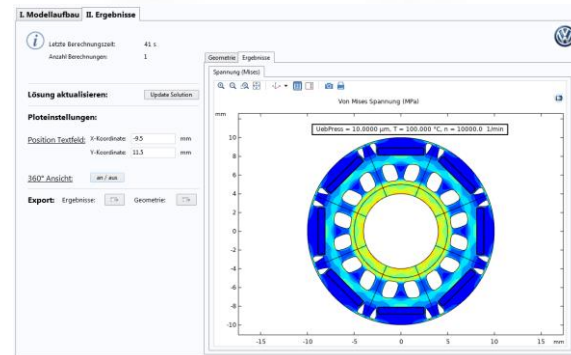
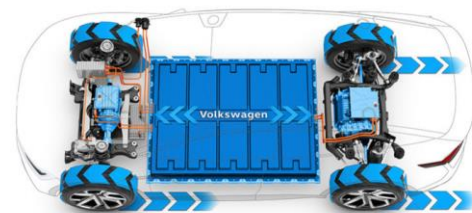
Electric Vehicle Motor Development with Apps

Volkswagen Kassel accelerates the design process for electric motors by building and distributing simulation applications that evaluate the strength of rotor laminations.

The VW Kassel plant produces 150 electric and 300 hybrid drives daily. Balancing dual requirements, such as torque, or power, with durability can be challenging.



The simulation experts model electric drive designs to address requirements early on in the development process, greatly reducing the number of variants. From there, they build simulation applications.



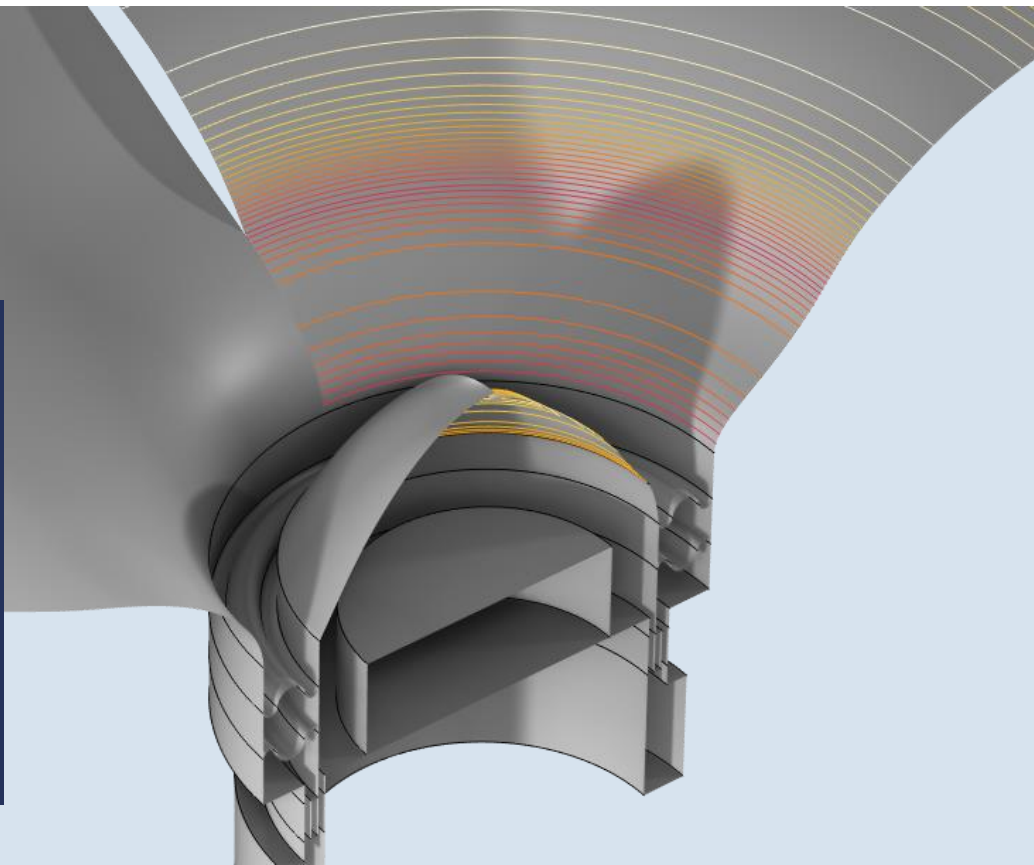
By using a simulation app based on the model, designers can quickly and easily benchmark different variants.

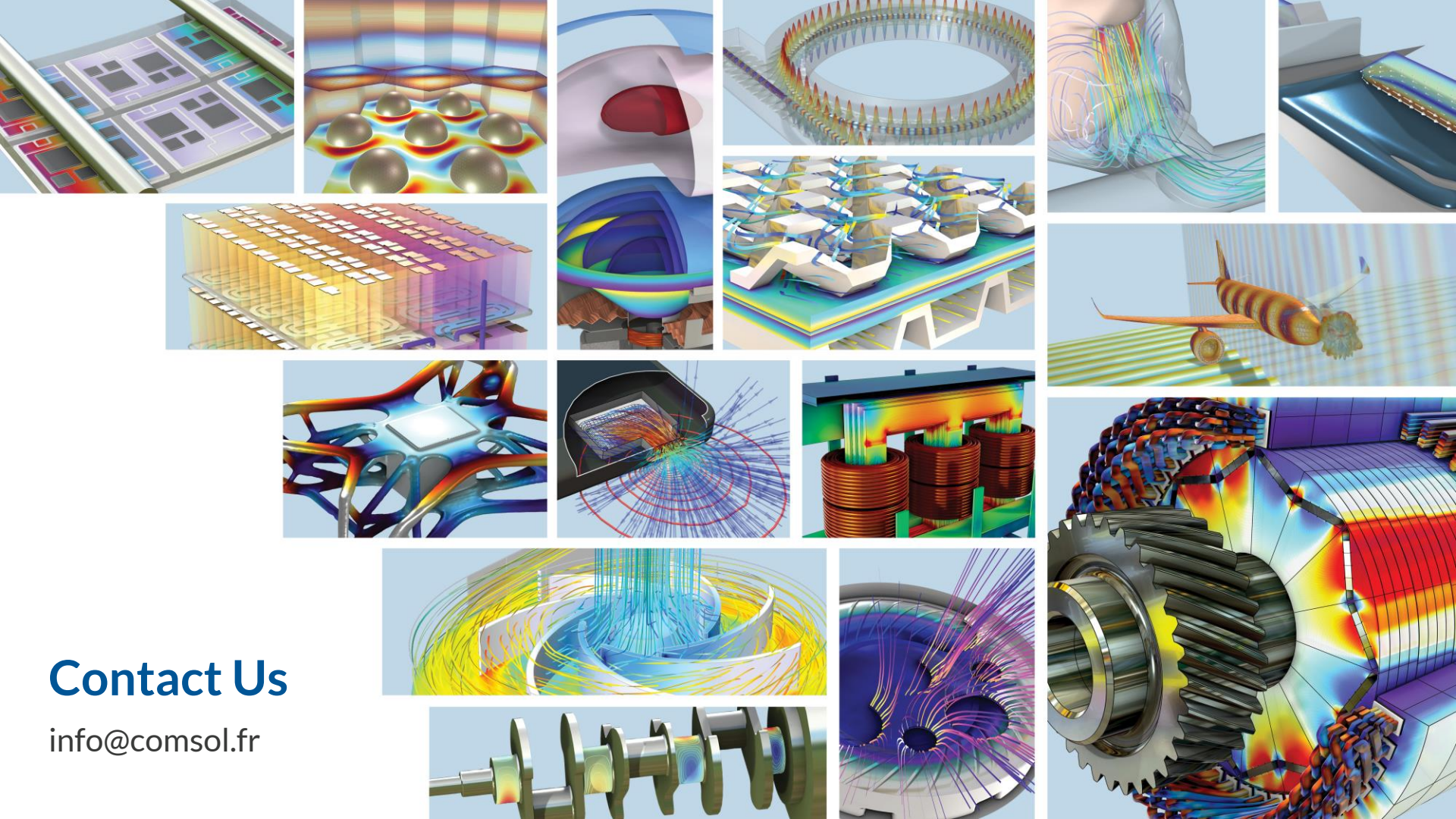
The test process for evaluating the strength of rotor laminations has been automated through simulation apps — further reducing development costs and increasing product quality.

CONCLUSION

Improve Existing Designs

Shape and topology optimization can improve existing designs by combining user-defined objectives with the built-in functionality in COMSOL Multiphysics®.





Contact Us

info@comsol.fr